Decoding Motor Intent from sEMG Signals

Machine Learning for Hand Gesture Classification

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42-675 Fundamentals of Computational BME

Introduction: Medical Background

Neuromuscular Physiology

Human movement requires precise brain-spinal cord-muscle communication. When disrupted by SCI or stroke, neural transmission is reduced but not eliminated—many patients retain weak, measurable electrical activity in affected muscles.

Surface EMG as a Bridge

sEMG non-invasively records residual motor unit activation through skin electrodes. Each gesture recruits unique muscle activation patterns, enabling motor intent detection even with limited visible movement.

Machine Learning Potential

Traditional methods struggle with complex, non-linear EMG patterns. ML can decode these signals effectively, translating residual activity into control commands for prosthetics and rehabilitation systems.

[1]Atzori et al., 2014; Phinyomark & Scheme, 2018

Introduction: Goals & Existing Work

Project Goal

Develop and compare ML models for decoding hand gestures from sEMG signals to advance prosthetic control and rehabilitation interfaces

Significance

- Non-invasive restoration of control
- Natural, intuitive interaction
- Improved independence for patients
- Applicable to prosthetics, exoskeletons, HCI

Existing ML Approaches

Hudgins et al. (1993)[2]

Established time-domain features (TD set) with neural networks for multi-function control

Englehart & Hudgins (2003)[3]

Real-time LDA controller with minimal latency—robust for online myoelectric control

Our Contribution

Compare classical LDA approach with modern Neural Networks for automatic feature learning; evaluate cross-day generalization

[2] Hudgins et al. 1193; [3] Englehart & Hudgins, 2003

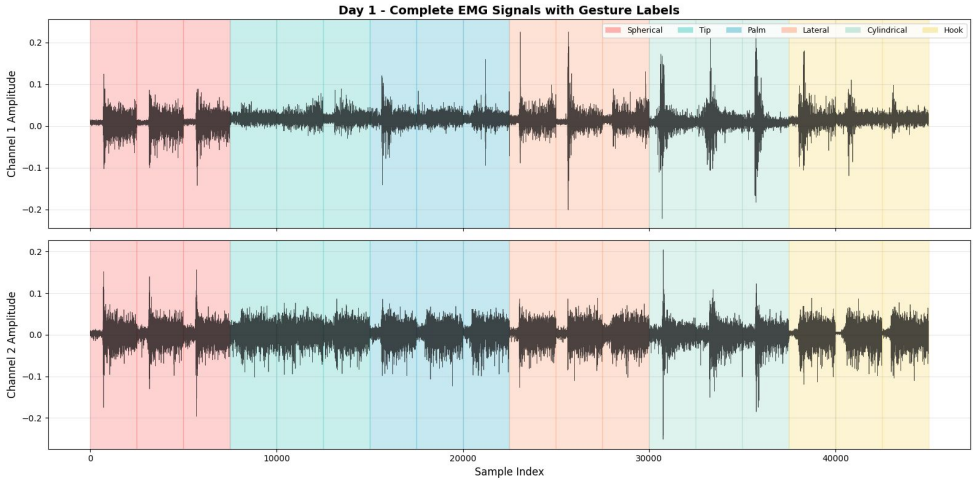
Methods: Dataset Summary

Data Source[4]

UCI ML Repository: *sEMG for Basic Hand Movements* (Sapsanis et al., 2013)

Database

- 1 subject, 3 days, Sampling rate: 500 Hz
- 6 gestures x 100 reps/day
- 5 seconds/trial
- **1,800 total trials**

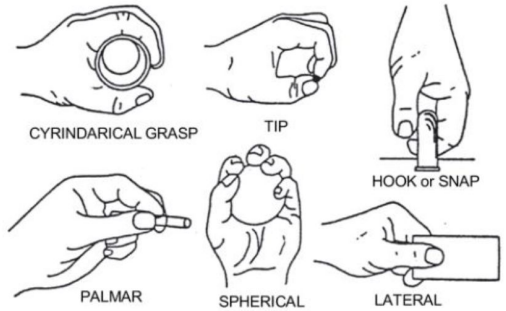


Recording Details

Two Channels

- Flexor Carpi Ulnaris
- Extensor Carpi Radialis

Six Gesture Classes



Balanced dataset: 100 trials per gesture per day
Total: ~8M samples per channel

Methods: Data Preprocessing

1. **Filtering (was performed by database authors):** Bandpass (15-500 Hz) + Notch (50 Hz) to remove power line interference
2. **Segmentation:** Sliding windows (100 sample windows for LDA; 128 samples for CNN) with 50% overlap
3. **Rectification (LDA & MLP):** Full-wave rectification
4. **Normalization:** L2 norm per trial per window to handle amplitude variations
5. **Feature Extraction (LDA & MLP):** Time-domain features from rectified, smoothed signals
6. **Standardization (LDA & MLP):** Z-normalization using StandardScaler applied to extracted features

LDA Approach:

Manual feature extraction → 2 channels × 10 features (RMS, MAV, ZC, SSC, WL) → StandardScaler (Z-normalization) (mean=0, std=1) → PCA (95% variance)

CNN Approach:

Raw normalized windows → Automatic hierarchical feature learning through convolution

Methods: Evaluation Metrics

Within-Day Evaluation
- Separate model per day
- 80% train / 20% test split
- Tests same-session performance

Cross-Day Evaluation
- Train on Day 1 (80%)
- Test on Days 1, 2, 3 (20% each)
- Tests generalization across sessions

Stratified Sampling
Maintains class balance in train/test splits (100 trials per gesture per day)

Overall Accuracy

Percentage of correctly classified gestures across all classes

Per-Class Metrics

- **Precision:** % of predicted class that are correct
- **Recall:** % of actual class correctly identified
- **F1-Score:** Harmonic mean of precision & recall

Confusion Matrix

Visualizes misclassification patterns between gesture pairs (e.g., Tip vs. Lateral)

Methods: Features

A **Features Extracted From Smoothed Signal**

$$\mathrm{RMS}_i = \sqrt{\frac{1}{L}\sum_{n=1}^{L}(x_i[n])^2} \qquad \mathrm{MAV}_i = \frac{1}{L}\sum_{n=1}^{L}|x_i[n]| \qquad \mathrm{WL}_i = \sum_{n=1}^{L-1}|x_i[n+1]-x_i[n]|$$

B **Feature Extracted From Filtered Signal**

$$\mathrm{ZC}_i = \frac{1}{2}\sum_{n=1}^{L-1}|\mathrm{sgn}(x_i[n]) - \mathrm{sgn}(x_i[n+1])|$$

C **Feature Extracted From Rectified Signal**

$$\mathrm{SSC}_i = \sum_{n=2}^{L-1}\mathbb{1}[(x_i[n]-x_i[n-1])(x_i[n]-x_i[n+1])>0 \text{ and } |x_i[n+1]-2x_i[n]+x_i[n-1]|>\theta]$$

Input Features (10 total)

Per channel, per 100 sample window:

- RMS (smoothed, rectified)
- MAV (smoothed, rectified)
- ZC (raw signal)
- SSC (rectified signal)
- WL (smoothed, rectified)

→ PCA dimensionality reduction (95% variance)

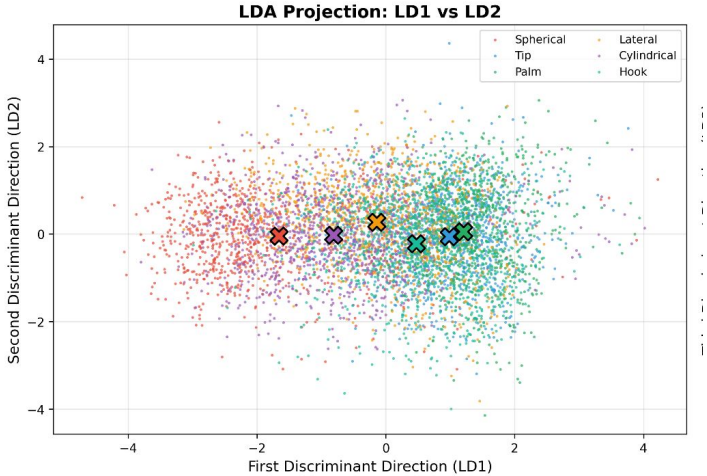
Methods: Algorithm 1 - Linear Discriminant Analysis (LDA)

Algorithm Description

Principle: Linear classifier maximizes class separation

Assumptions:

- Gaussian class distribution
- Shared covariance matrix
- Linear separability



Implementation Details

Hyperparameters:

- Solver: SVD
- Shrinkage: None
- Default class priors

Data Split:

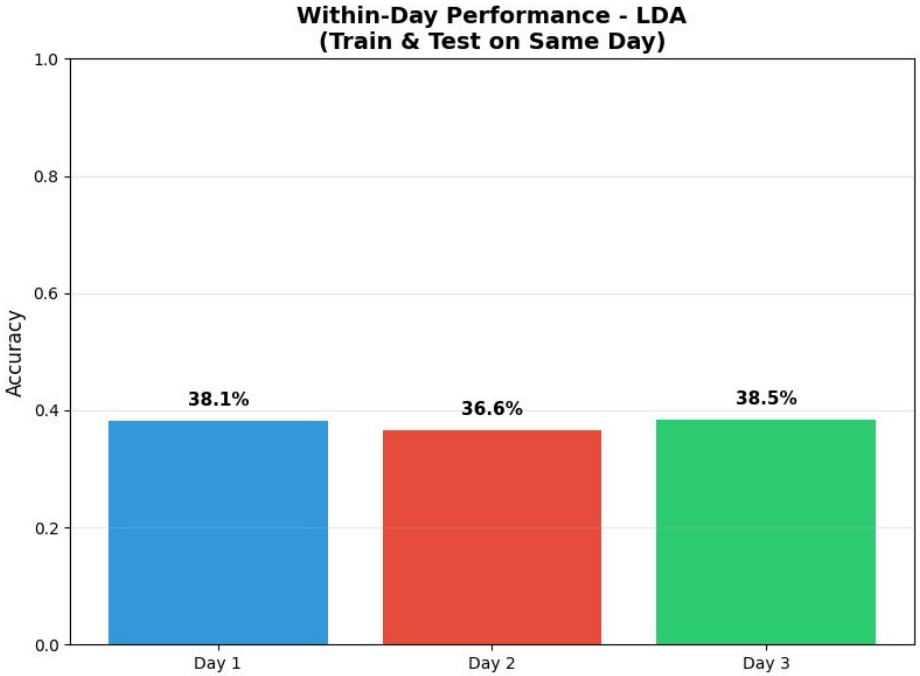
- Within-day: 80% train, 20% test (per day)
- Cross-day: Train: Day 1, Test: Days 1-3

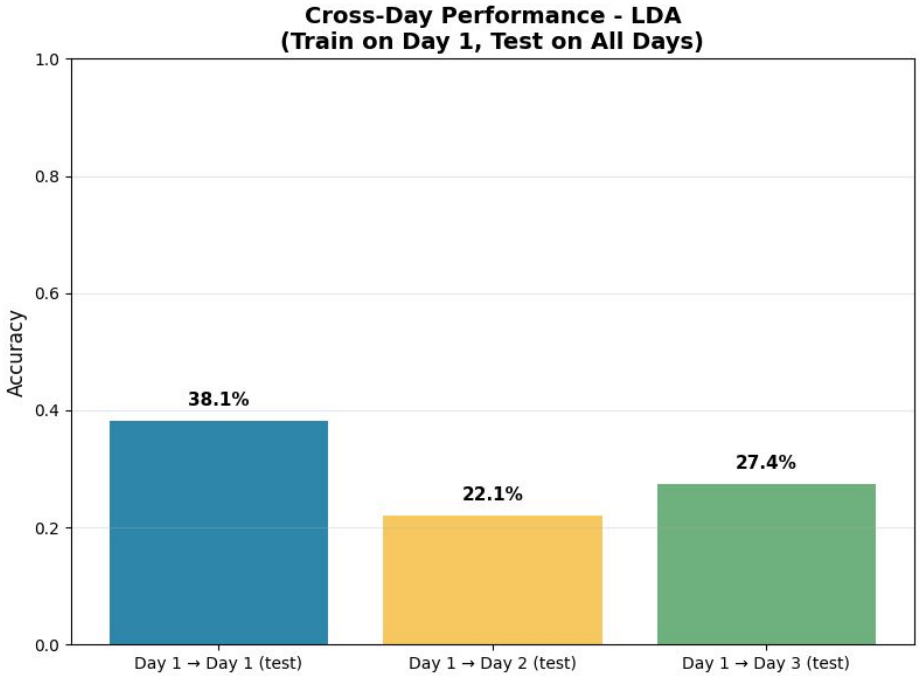
Advantages:

- Fast training and prediction
- Interpretable linear weight
- Low computational cost
- Suitable for real-time control

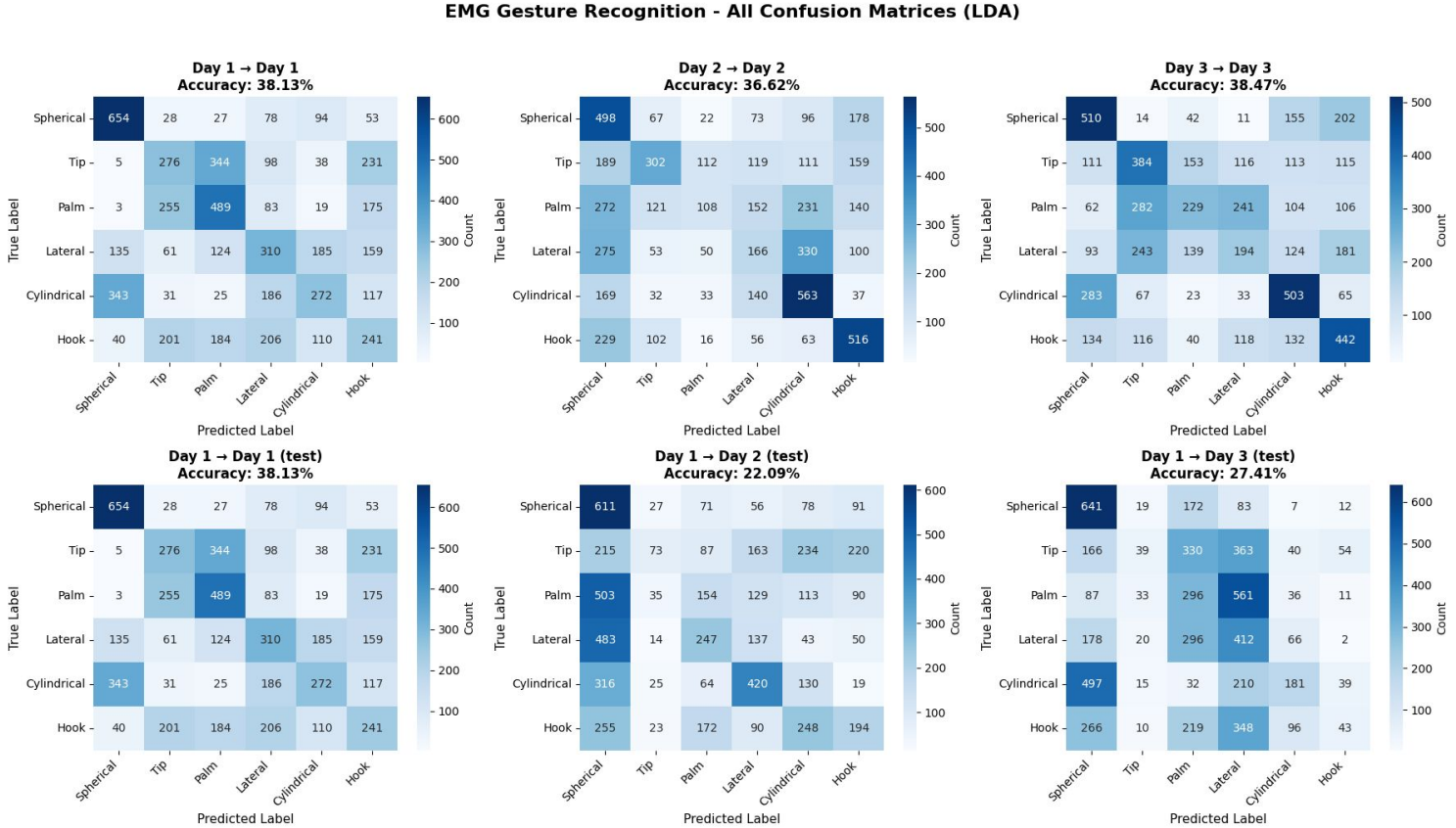
[2] Hudgins et al. 1193; [3] Englehart & Hudgins, 2003

Results - LDA


Within-Day Performance - LDA
(Train & Test on Same Day)
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
38.1%
36.6%
38.5%
Day 1
Day 2
Day 3


Cross-Day Performance - LDA
(Train on Day 1, Test on All Days)
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
38.1%
22.1%
27.4%
Day 1 → Day 1 (test)
Day 1 → Day 2 (test)
Day 1 → Day 3 (test)

Results - LDA


EMG Gesture Recognition - All Confusion Matrices (LDA)
Day 1 → Day 1
Accuracy: 38.13%
Day 2 → Day 2
Accuracy: 36.62%
Day 3 → Day 3
Accuracy: 38.47%
Day 1 → Day 1 (test)
Accuracy: 38.13%
Day 1 → Day 2 (test)
Accuracy: 22.09%
Day 1 → Day 3 (test)
Accuracy: 27.41%
True Label
Predicted Label
Count
Spherical
Tip
Palm
Lateral
Cylindrical
Hook

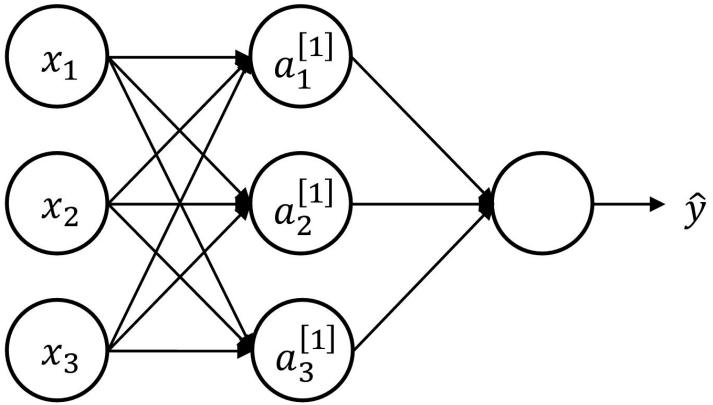
Methods: Algorithm 2 - MLP

Algorithm Description

Principle: Combination of linear perceptrons which when stacked, can predict non-linear data.

Overview:

- Forward propagation predicts an output and calculates error
- Back propagation uses that error in logistic regression to update the weights



Implementation Details

Hyperparameters:

- # of Nodes per Layers
- # of Hidden Layers
- Initialized weights

Data Split:

- Within-day: 80% train, 20% test (per day)
- Cross-day: Train: Day 1, Test: Days 1-3

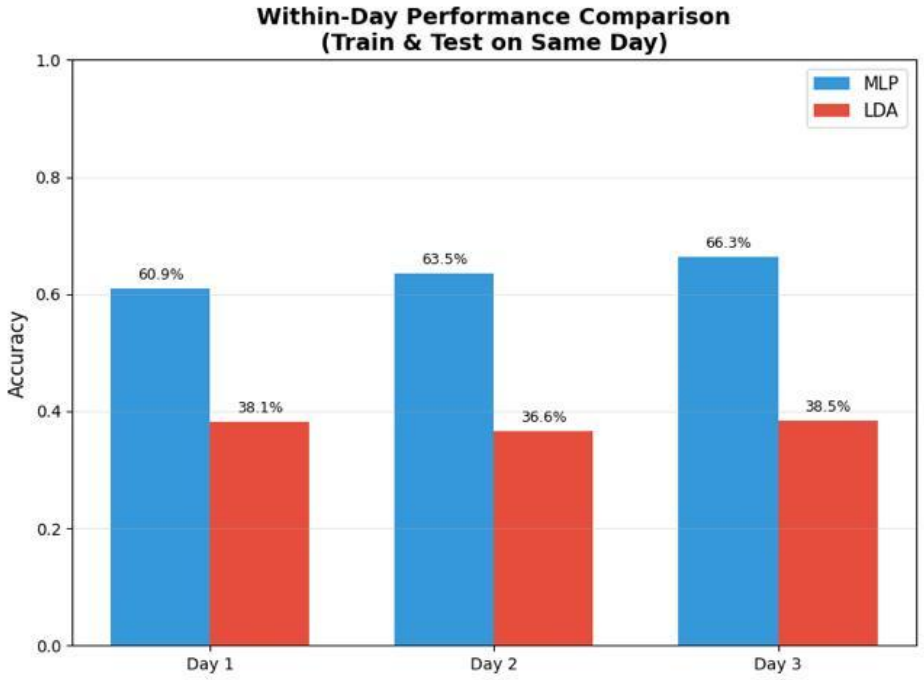
Advantages:

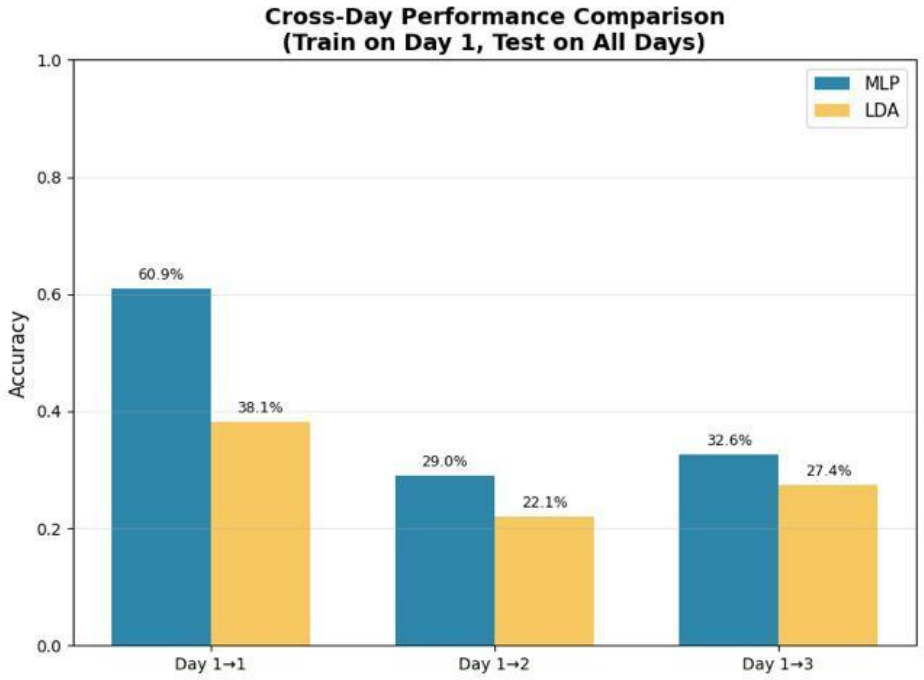
- Learns optimal features automatically
- Can estimate non-linear decision boundaries

Results - MLP


EMG Gesture Recognition - All Confusion Matrices (MLP)
Day 1 → Day 1
Accuracy: 60.85%
Day 2 → Day 2
Accuracy: 63.49%
Day 3 → Day 3
Accuracy: 66.33%
Day 1 → Day 1 (test)
Accuracy: 60.85%
Day 1 → Day 2 (test)
Accuracy: 29.01%
Day 1 → Day 3 (test)
Accuracy: 32.64%
True Label
Predicted Label
Count
Spherical
Tip
Palm
Lateral
Cylindrical
Hook

Results - MLP


Within-Day Performance Comparison
(Train & Test on Same Day)
MLP
LDA
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
60.9%
38.1%
63.5%
36.6%
66.3%
38.5%
Day 1
Day 2
Day 3


Cross-Day Performance Comparison
(Train on Day 1, Test on All Days)
MLP
LDA
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
60.9%
38.1%
29.0%
22.1%
32.6%
27.4%
Day 1→1
Day 1→2
Day 1→3

Results

Cross Day Calibration	Accuracy		F1 Score	
	LDA	MLP	LDA	MLP
1	38.13%	60.85%	37.31%	60.53%
2	22.09%	29.01%	19.48%	24.94%
3	27.41%	32.64%	23.04%	28.03%

Within Day Calibration	Accuracy		F1 Score	
	LDA	MLP	LDA	MLP
Day 1	38.13%	60.85%	37.31%	60.53%
Day 2	36.62%	63.49%	34.34%	62.09%
Day 3	38.47%	66.33%	37.71%	66.00%

Discussion: Model Comparison & Selection

Criterion	LDA	MLP
Within-Day Accuracy	~36-39%	~60-67%
Cross-Day Stability	Moderate degradation (~18%)	Significant degradation (~ 35%)
Training Time	Fast (seconds)	Medium
Feature Engineering	Manual (domain expertise)	Automatic Learning
Interpretability	High (linear weights)	Low (black box)
Nonlinear Patterns	Limited capacity	Excellent

Discussion and Conclusion

Limitations

Poor cross-day performance - models don't generalize after Day 1
Electrode placement sensitivity - signal patterns change when sensors move
Daily re-training required - good accuracy only when trained fresh each day
Feature drift in EMG signals - muscle activation patterns shift across sessions

Benefits

High same-day accuracy - especially with MLP
Real-time capable models - fast inference for prosthetic control
Automatic feature learning - MLP
Removes need for manual feature design

Future Directions

Domain adaptation - handle signal drift without full retraining
Online calibration - quick personalized recalibration each session
Hybrid features - combine learned + handcrafted features
More channels / robust sensors - improve spatial stability
Better Models like using, CNN or Transfer Learning

References

[1] Hudgins, B., Parker, P. A., & Scott, R. N. (1993). A new strategy for multifunction myoelectric control. IEEE Transactions on Biomedical Engineering, 40(1), 82–94. https://doi.org/10.1109/10.204774

[2] Englehart, K., & Hudgins, B. (2003). A robust, real-time control scheme for multifunction myoelectric control. IEEE Transactions on Biomedical Engineering, 50(7), 848–854. https://doi.org/10.1109/TBME.2003.813539

[3] Sapsanis, C., Georgoulas, G., Tzes, A., & Lymberopoulos, D. (2013). Improving EMG based classification of basic hand movements using EMD. 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 5754–5757. https://doi.org/10.1109/EMBC.2013.6610858

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[5] Atzori, M., Gijsberts, A., Castellini, C., Caputo, B., Mittaz Hager, A.-G., Elsig, S., Giatsidis, G., Bassetto, F., & Müller, H. (2014). Electromyography data for non-invasive naturally-controlled robotic hand prostheses. Scientific Data, 1, 140053. https://doi.org/10.1038/sdata.2014.53

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